

A non-local algorithm for image denoising

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a 2005 paper by
Antoni Buades, Bartomeu Coll, Jean-Michel Morel

INTRODUCTION

to image denoising

What is “image denoising”?

A digital image is, in essence, a matrix.

Preceding Research

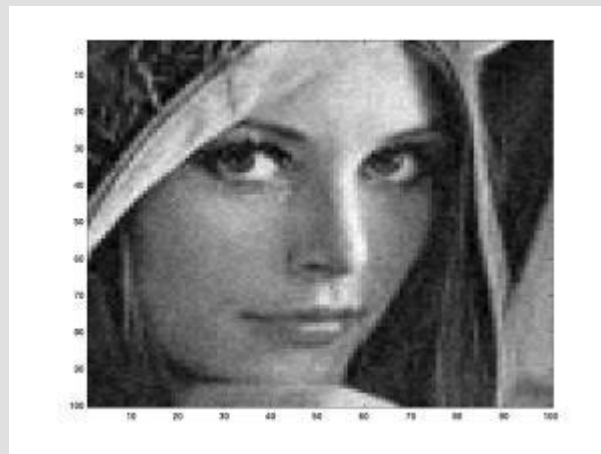
Buades, Coll, and Morel had written two papers together prior to this one.

1. *On image denoising methods*. Technical Report 2004-15, CMLA, 2004.
2. *Neighborhood filters and pde 's*. Technical Report 2005-04, CMLA, 2005.

“Two main limitations in image accuracy are categorized as **blur** and **noise**.”

from *On image denoising methods*.

Original Image vs. Noisy Image



“The goal of image denoising is to **recover the original image from a noisy measurement.**”

$$v(i) = u(i) + n(i)$$

observed value

"true" value

noise perturbation

Define a **denoising method** $\mathbf{D}_h$ as

$$\mathbf{v} = \mathbf{D}_h \mathbf{v} + \mathbf{n}(\mathbf{D}_h, \mathbf{v})$$

How do we remove noise from images?

Denoising is achieved by averaging.

How do we measure the accuracy
of denoising methods?

The Method Noise

$$u - D_h u$$

LOCAL SMOOTHING FILTERS

and their image method noises

The Gaussian Filtering

$$u - G_h * u = -h^2 \Delta u + o(h^2)$$



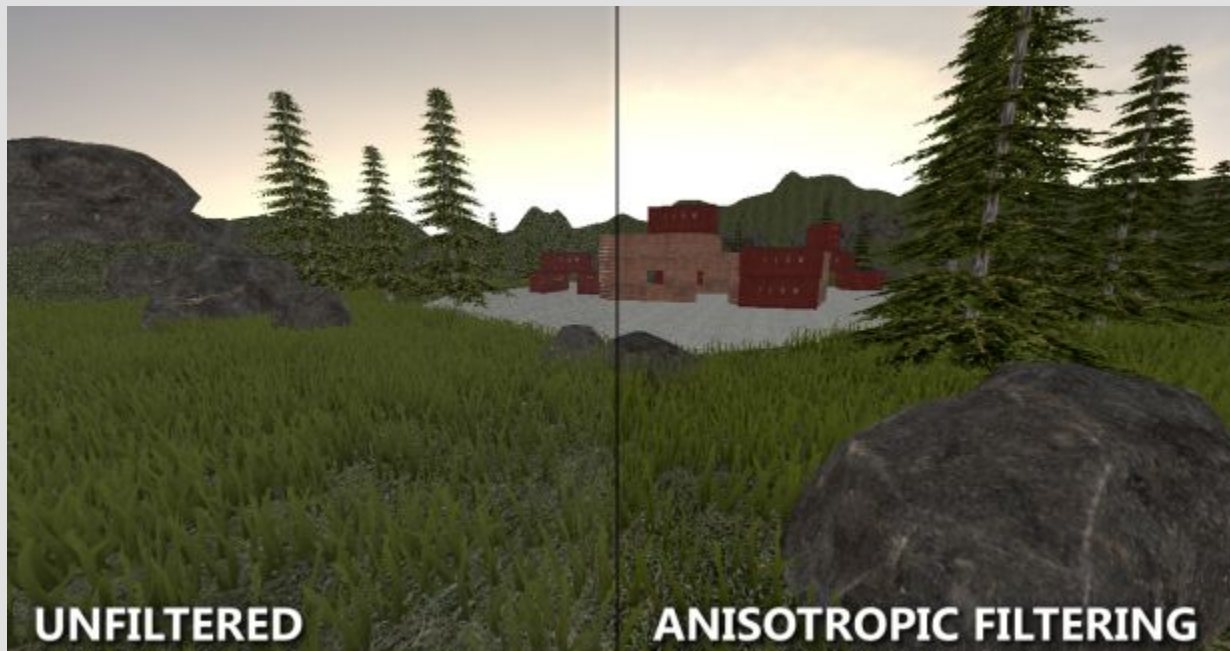
Noisy input: PSNR = 39.1 dB



Gaussian filtered: PSNR = 67.9 dB

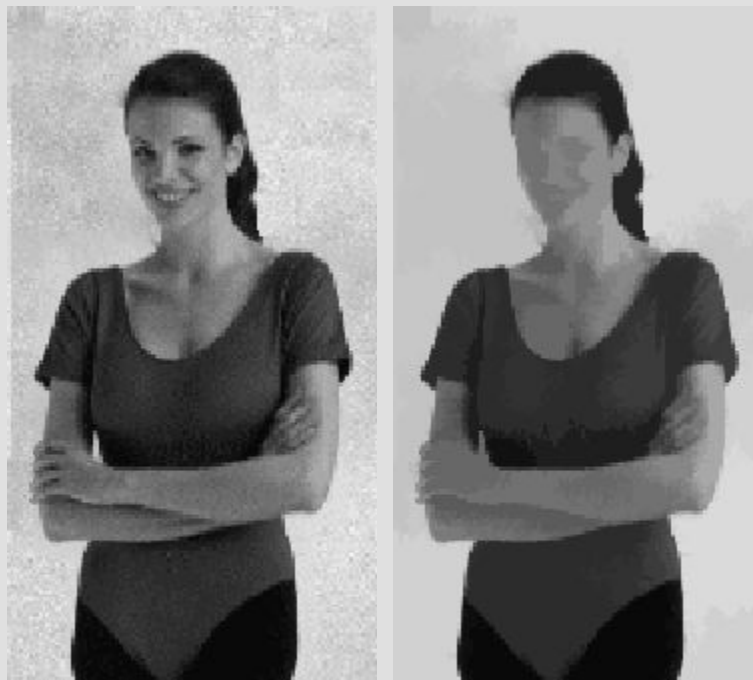
The Anisotropic Filtering

$$u(x) - AF_h * u(x) = -\frac{1}{2}h^2 |Du| \text{curv}(u)(x) + o(h^2)$$



The Total Variation Minimization

$$u(x) - \text{TVF}_\lambda(u)(x) = -1/(2\lambda)\text{curv}(\text{TVF}_\lambda(u))(x)$$



The Neighborhood Filtering

$$YNF_{h,p} u(x) = \frac{1}{C(x)} * \int_{B_p(x)} u(y) * e^{-\frac{|u(y)-u(x)|^2}{h^2}} d(y)$$

Yaroslavsky filter averages pixels with similar grey level value and belonging to the spatial neighborhood $B_p(\mathbf{x})$

The NL-Means Algorithm

The Non-Local Means Algorithm

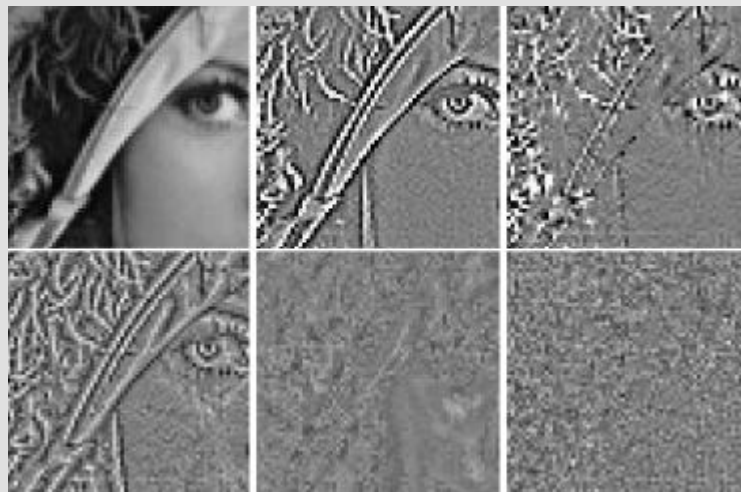


Weighted average of all
pixel neighborhoods
throughout the image

Main points



NL-Means

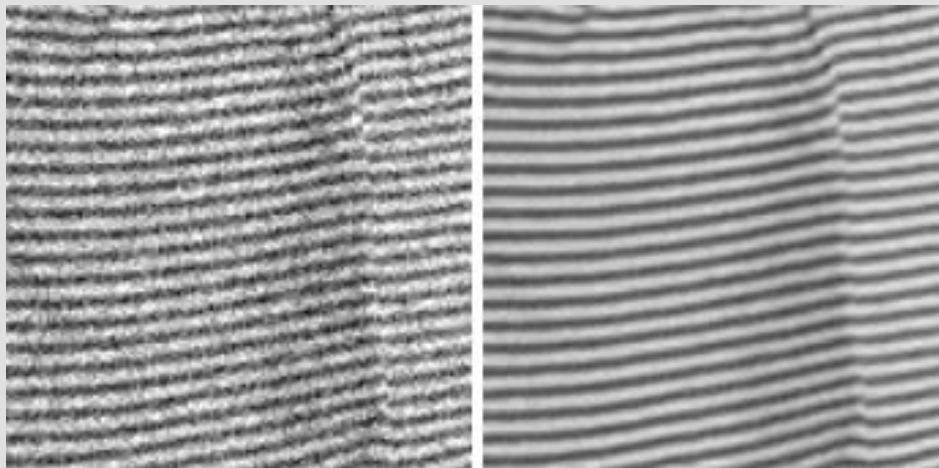


NL-Means

Image	σ	GF	AF	TV	TV Iter	YNF	EWf	TIHWT	NL-means
Boat	8	53	38	39	29	39	33	28	23
Lena	20	120	114	110	82	129	105	81	68
Barbara	25	220	216	186	189	176	111	135	72
Baboon	35	507	418	365	364	381	396	365	292
Wall	35	580	660	721	715	598	325	712	59

Strengths of the NL-Means Algorithm

- Functions well with periodicity of images, takes advantage of patterns
- Recognize similar patterns in far away pixels, even in natural images



Weaknesses of the NL-Means Method

- Loss of detail and clarity (look especially at the clouds and the mountain)
- All images have had noise added around a standard deviation, uniform at each pixel - often, images have areas with more noise in local areas



CONCLUSION

1. Method Noise
2. Local Smoothing Filters
3. NL-Means Algorithm

Assumptions

- Because of assumptions in the frequencies of images, sometimes fine details and the structure of the images are removed along with the noise
- Some of these methods, such as the NL Means Algorithm, assume redundancy in the images

Limitations

- Differentiating between textures and details vs noise
 - averaging cannot differentiate
- Ringing, Blur, Staircase Effect, Checkerboard
 - intrinsic in the algorithm model
 - assume white noise is oscillatory, and image is smooth, so smooths image and creates issues

Improvements

- Averaging the final denoised value of pixel 'i' with the value of $v(i)$ - thus placing it closer to the value of the noisy image
 - This will weaken the method noise model mathematically, but may compensate visually

Resources

Applied Math 120 - Applied Linear Algebra and Big Data (image processing)

Astronomy 193 - Noise and Data Analysis in Astrophysics (statistical analysis of data error)

Biomedical Engineering 128 - Intro to Biomedical Imaging and Systems (relates to class topic as well)

Computer Science 283 - Computer Vision - image segmentation and interaction

Brian Tracey and Joseph Noonan - Tufts University - Image processing

Thank you.



Questions?